

JPEG AI, Super-Resolution и их артефакты — проблемы внедрения нейросетевых алгоритмов

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Об авторе



- Заведущий Graphics&Media Lab ВМК МГУ и Video Analysis Lab института Искусственного Интеллекта МГУ
- Создатель сайтов по алгоритмам
 - <https://compression.ru/video>
 - <https://videoprocessing.ai>
 - <https://videoprocessing.github.io/>
- Области интересов: современное сжатие видео, измерение качества видео, четырехмерное видео
- Руководил 40+ проектами с компаниями **Intel, Cisco, Samsung, Huawei, Broadcom** и др.
- Автор №1* на [Habr.com](https://habr.com) в хабах «**AR и VR**», «**Искусственный интеллект**», «**Работа с видео**» и «**Видеотехника**»
- Сомневается в разумности Homo Sapiens

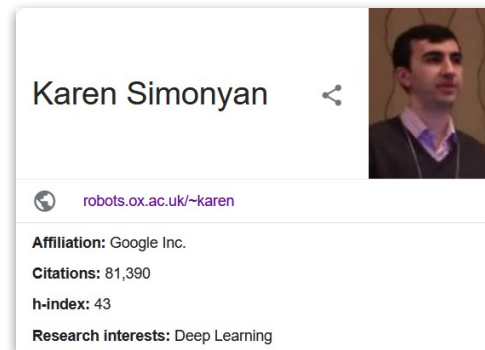
Our former student — author of VGG

Karen Simonyan is the first author of VGG — a revolutionary object-recognition model

VERY DEEP CONVOLUTIONAL NETWORKS
FOR LARGE-SCALE IMAGE RECOGNITION

Karen Simonyan* & **Andrew Zisserman***

Visual Geometry Group, Department of Engineering Science, University of Oxford
{karen,az}@robots.ox.ac.uk



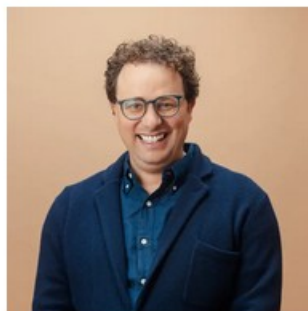
VirtualDub MSU Motion Estimation Filter

MSU Graphics & Media Lab (Video Group)

*Project, idea: Dr. Dmitry Vatolin
Algorithm: Karen Simonyan, Sergey Grishin
Implementation: Karen Simonyan*

TECH

The AI 100 2023: The top people in artificial intelligence

**Dario Amodei**

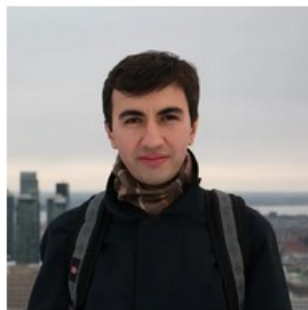
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Our worldwide leading results

#1 Video Codecs Analysis

compression.ru/video/codec_comparison

#1 Video Quality Estimation Tool

compression.ru/vqmt

#1 3D Video Quality Estimation metrics

videoprocessing.ai/stereo_quality

#1 The biggest website about data compression

compression.ru

#1 Video Processing Benchmarks Collection

videoprocessing.ai/benchmarks

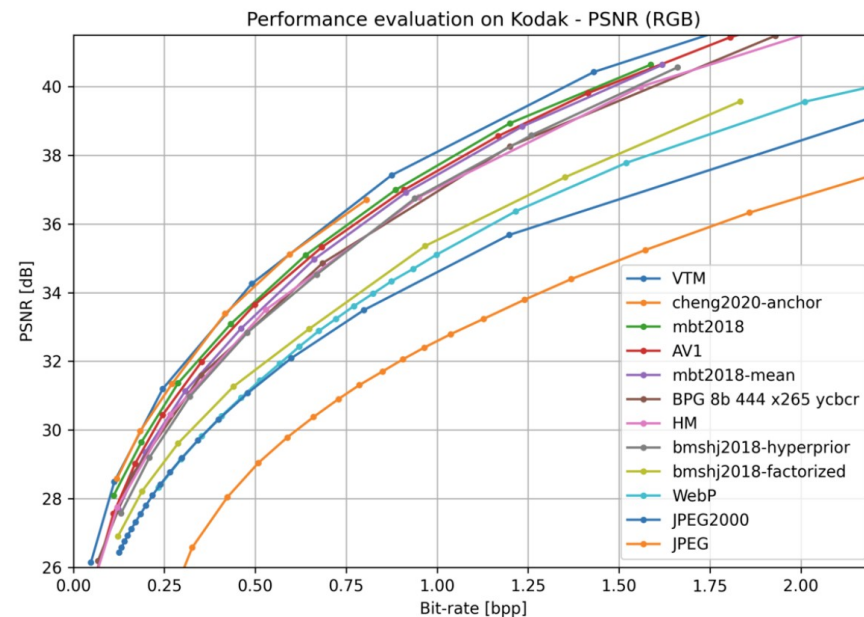
#1 globalcompetition.compression.ru (www.gdcc.tech) — world's biggest contest in lossless compression methods

Neural Artifacts by AI Compression

Introduction

Neural-network-based compression methods have been actively developed since 2018, catching up traditional approaches (HEVC, AV1 and VVC) in just **three years**

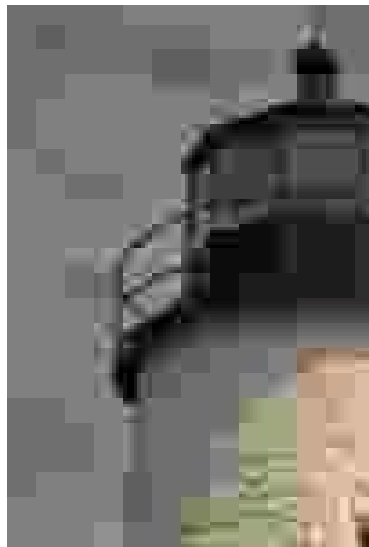
Methods **superior to VVC by 30%** in terms of image compression efficiency were proposed for the JPEG AI standard in 2022



Visual quality to compression ratio on the KODAK dataset

Advantages of neural-network-based compression methods

Neural networks can detect duplicate information in an image and convert it into a representation with less redundancy



JPEG, 103.2 times



JPEG2000, 111
times



MBT2018, 111.6
times

Advantages of neural compression methods

Neural networks can convert images into a representation with less redundancy



JPEG, 81.5 times



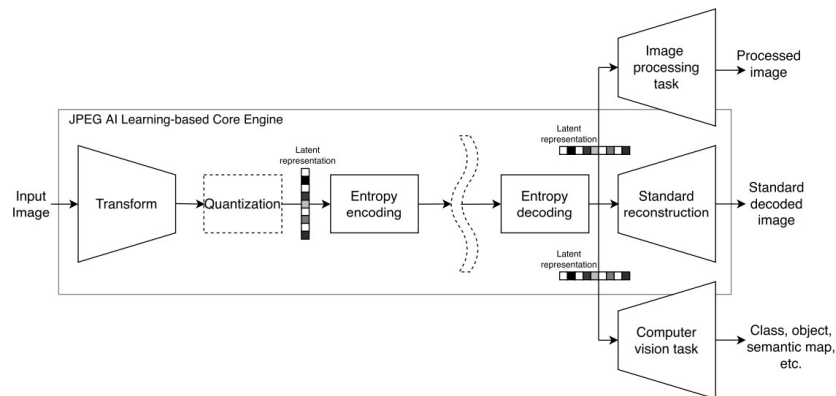
JPEG 2000, 83.7 times



JPEG AI, 80.2 times

JPEG AI

- Preliminarily: **release in 2024**
- Approximately **3 times more efficient than JPEG** (and 2 times more efficient than JPEG 2000)
- Challenges of the standard:
 - Targeting human and machine visions
 - Network size
 - **Quality metrics**
 - Computational complexity



JPEG AI framework scheme

Let's look into the
near future!

Texture distortion



Original



HM, 145.8 times



JPEG AI 3.2, 145.8 times

Texture distortion



Original



HM, 80.8 times

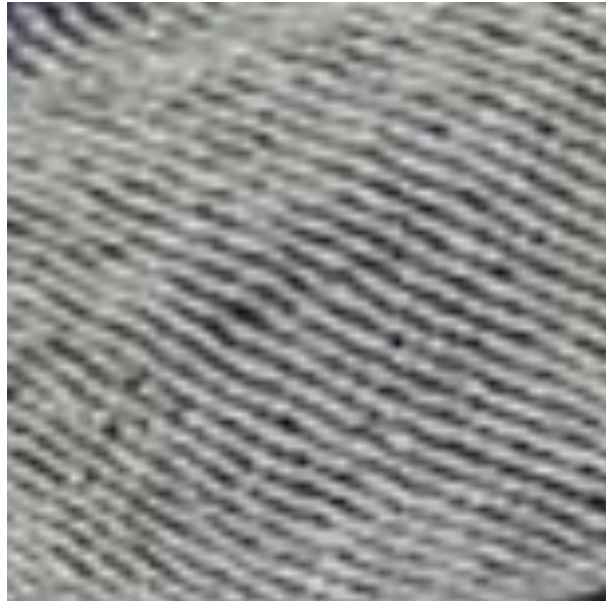


JPEG AI 3.2, 105.4 times

Texture distortion



Original



HM, 131.8 times



JPEG AI 4.1, 135.9 times

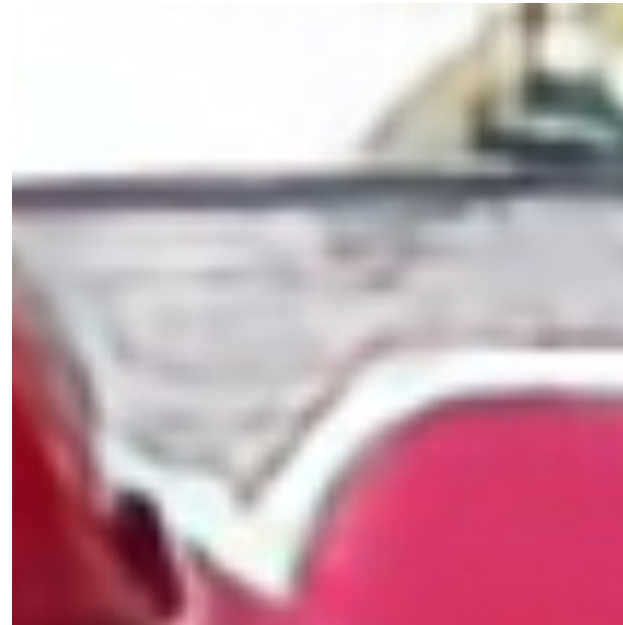
Texture distortion



Original

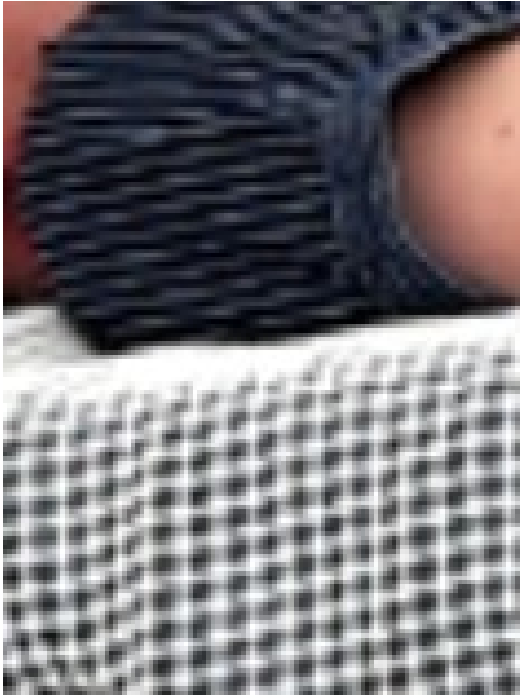


HM, 75.9 times

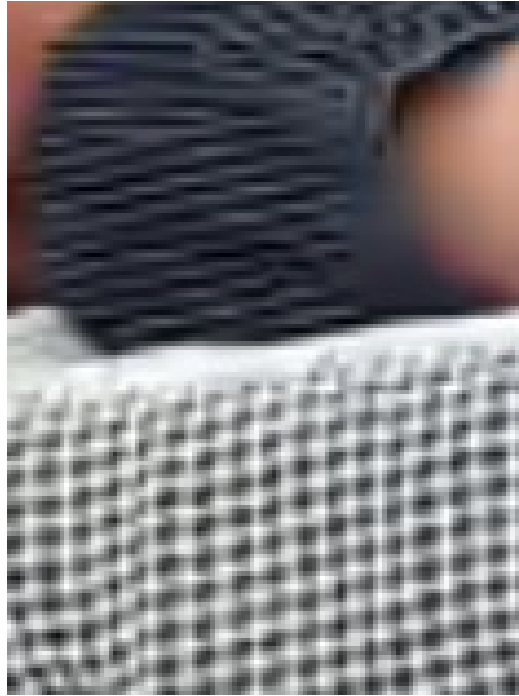


JPEG AI 4.1, 97 times

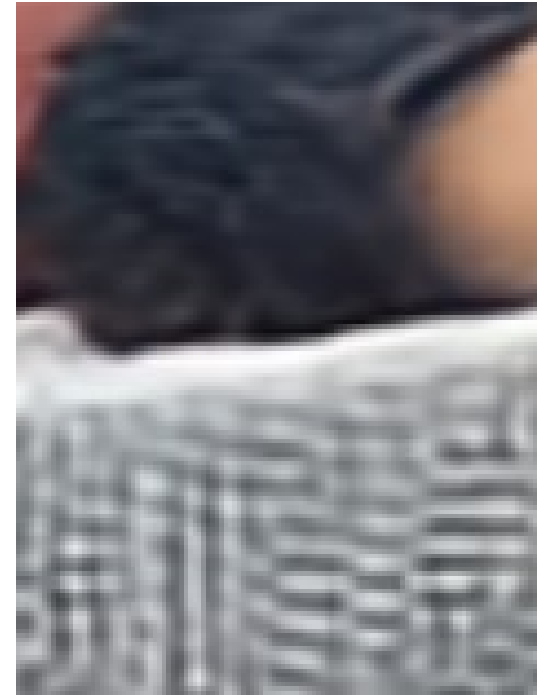
Texture distortion



Original



HM, 106 times

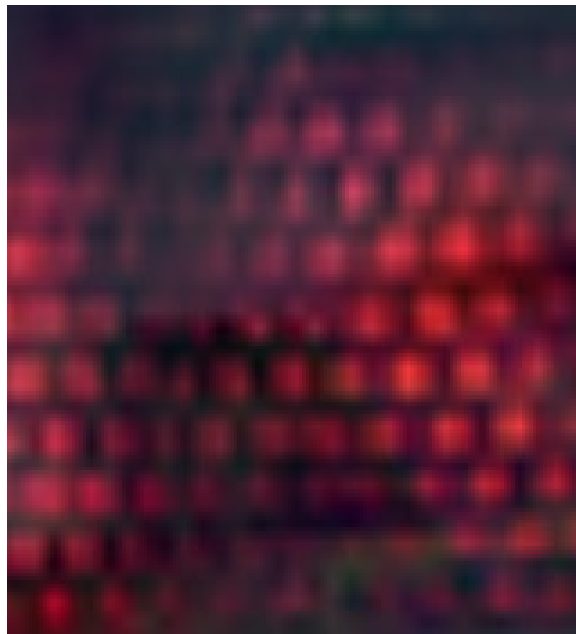


JPEG AI 4.1, 136 times

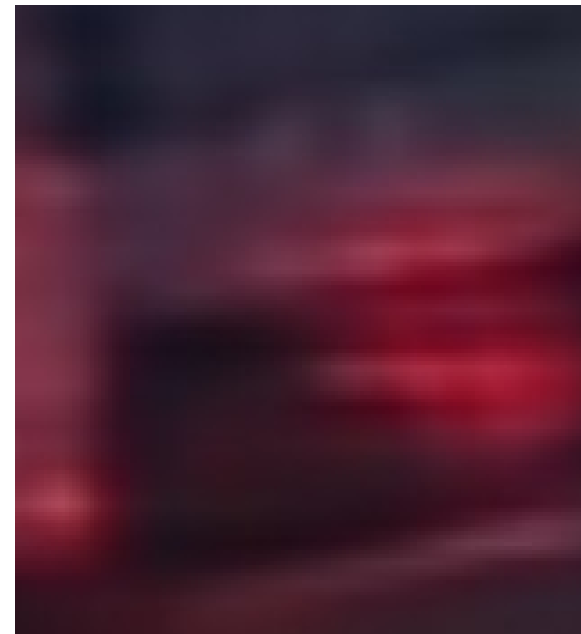
Texture distortion



Original



HM, 203.4 times



JPEG AI 4.1, 205.1 times

Texture distortion



Original



HM, 94.1 times



JPEG AI 4.1, 97.9 times

Texture distortion



HM, 123.2 times



JPEG AI 3.2, 128.4 times

Texture distortion

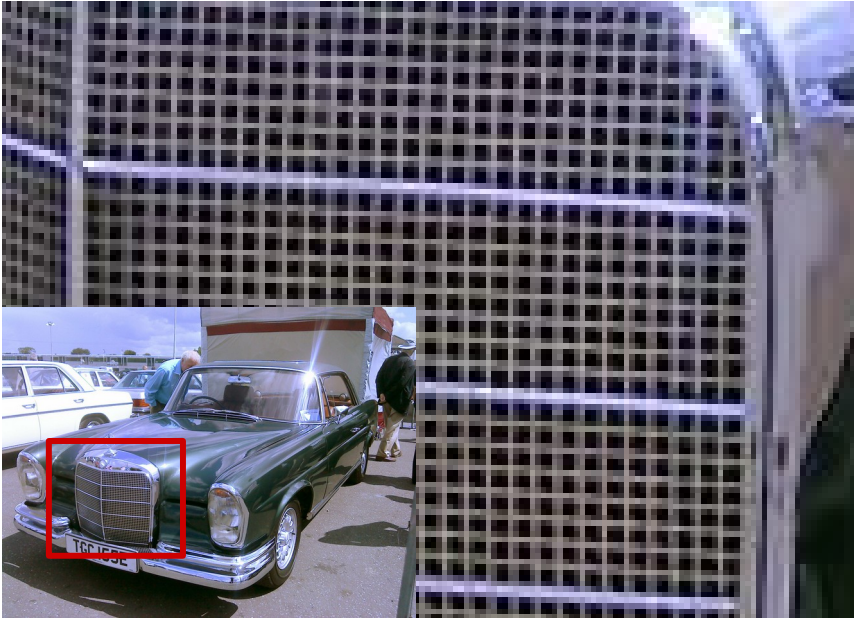


HM, 52.6 times



JPEG AI 3.2, 53.1 times

Texture distortion

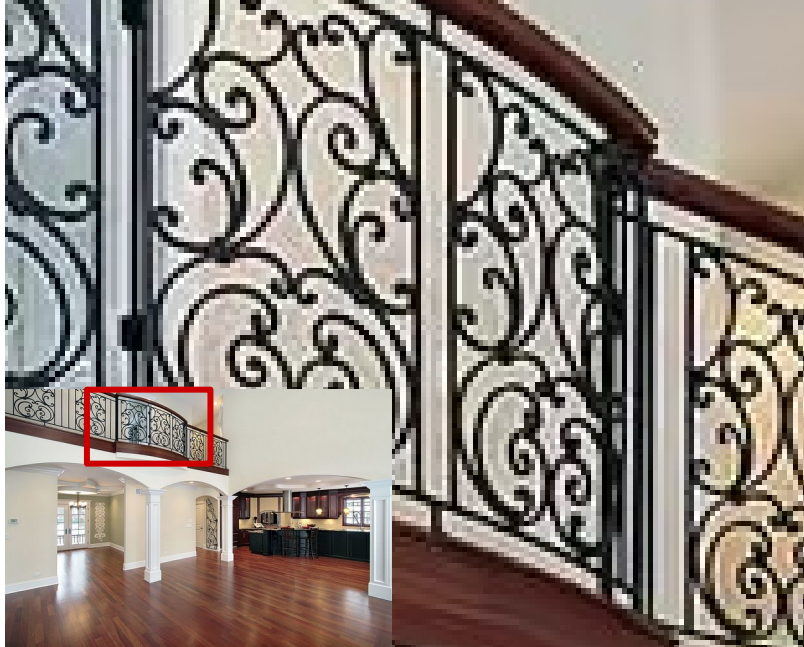


HM, 60.8 times



JPEG AI 4.1, 60 times

Texture distortion



HM, 77.1 times



JPEG AI 4.1, 79.2 times

Texture distortion



HM, 22.2 times



JPEG AI 4.1, 23.4 times

Color distortion

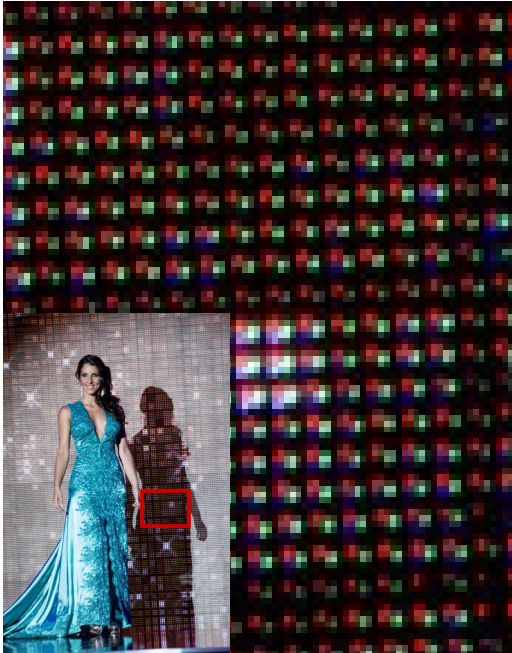


HM, 69.7 times



JPEG AI 3.2, 72.1 times

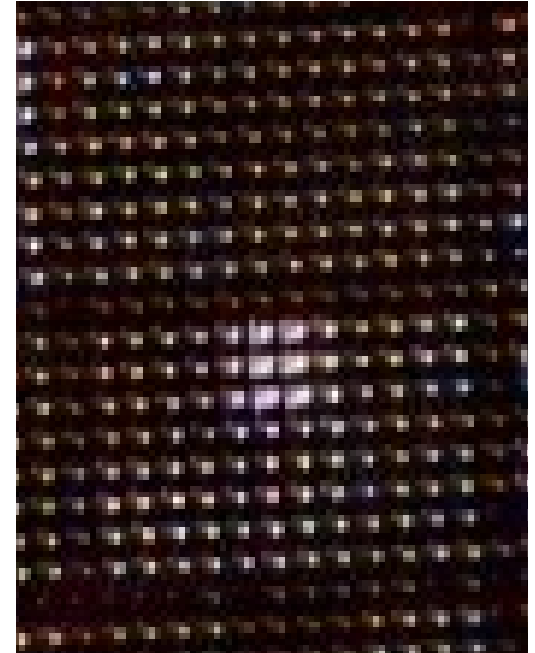
Color distortion



Original

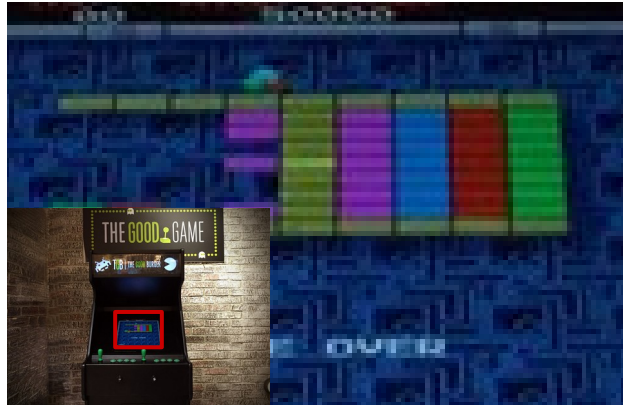


HM, 13.3 times



JPEG AI 4.1, 13.2 times

Color distortion



Original

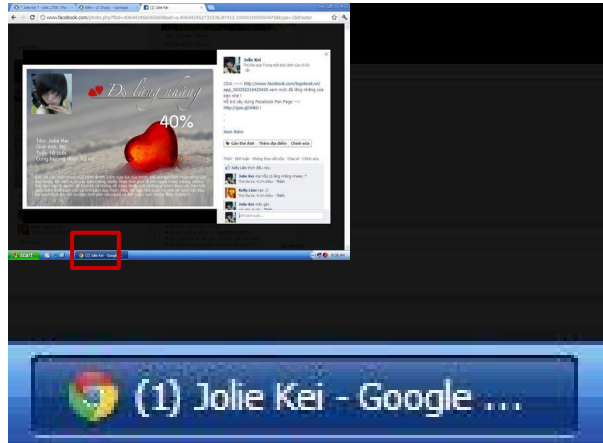


HM, 50.7 times

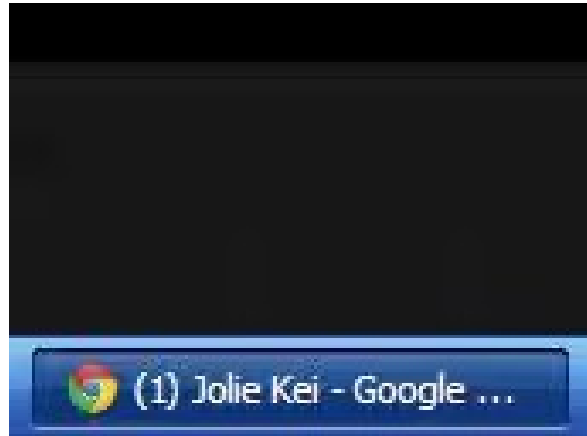


JPEG AI 4.1, 54.8 times

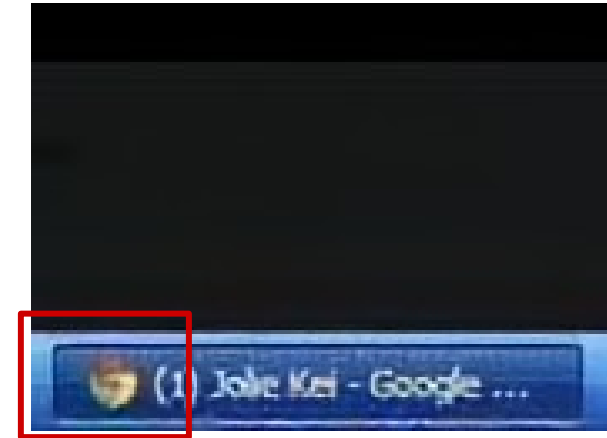
Color distortion



Original



HM, 74.8 times



JPEG AI 4.1, 77.8 times

Color and edges distortion



HM, 282 times



JPEG AI 3.2, 333 times

Edges distortion



HM, 22.2 times

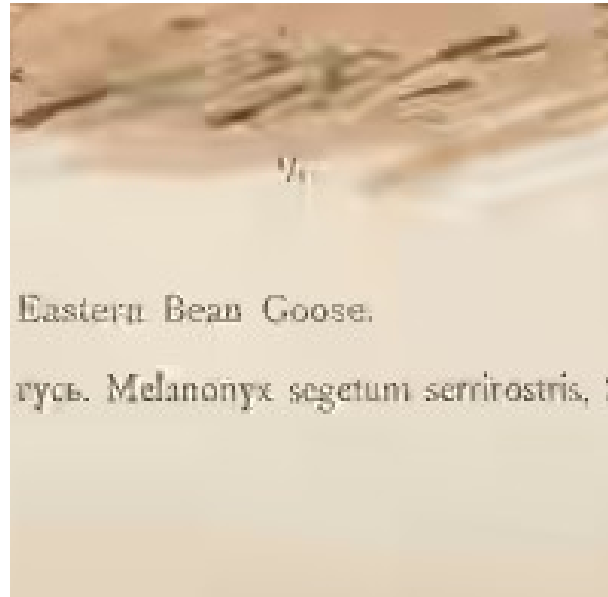


JPEG AI 4.1, 23.4 times

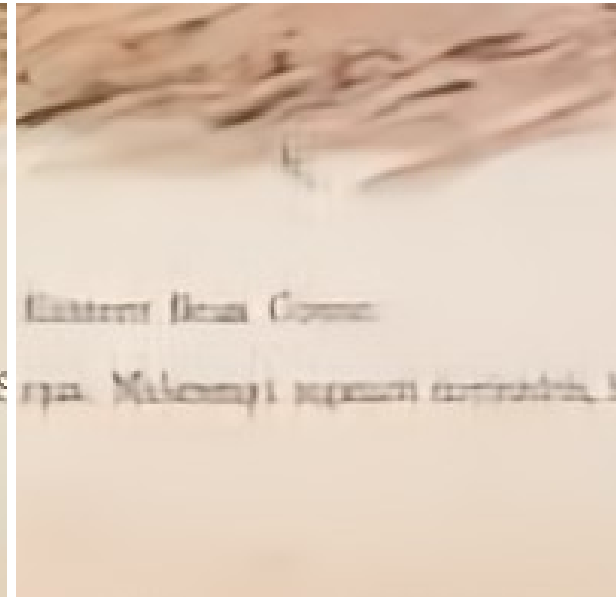
Text distortion



Original



HM, 363 times

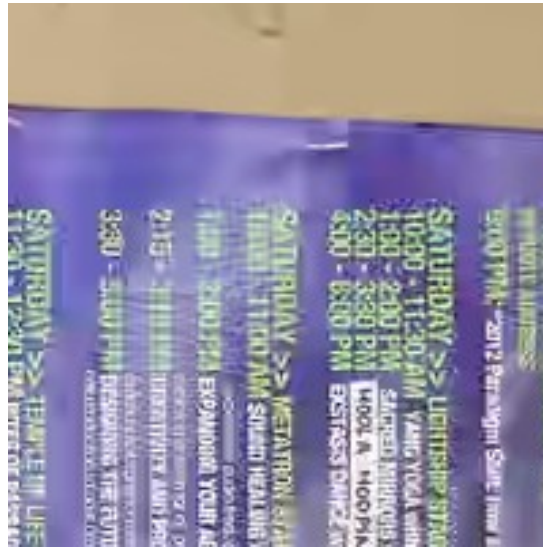


JPEG AI 3.2, 444 times

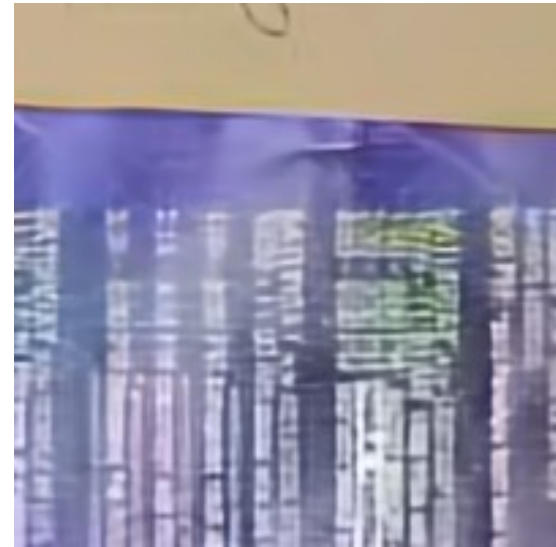
Text distortion



Original



HM, 152 times



JPEG AI 3.2, 190 times

Text distortion



Original



HM, 252 times



JPEG AI 3.2, 303 times

Text distortion



Original



HM, 196 times



JPEG AI 3.2, 230 times

Text and color distortion



Original



HM, 247 times



JPEG AI 3.2, 296 times

Text distortion



Original



HM, 200 times



JPEG AI 3.2, 240 times

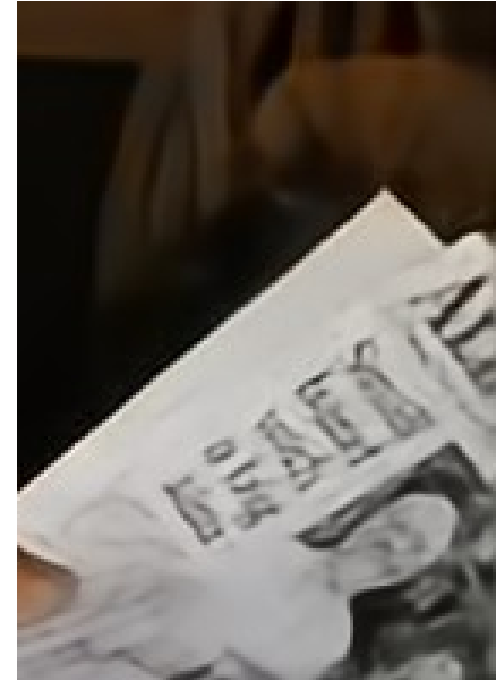
Text distortion



Original



HM, 289 times

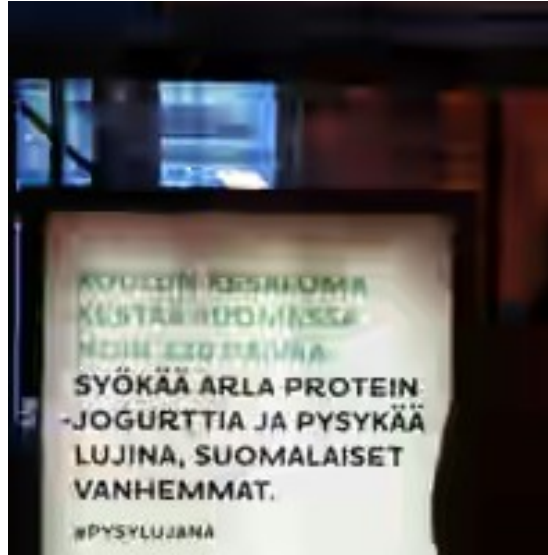


JPEG AI 3.2, 363 times

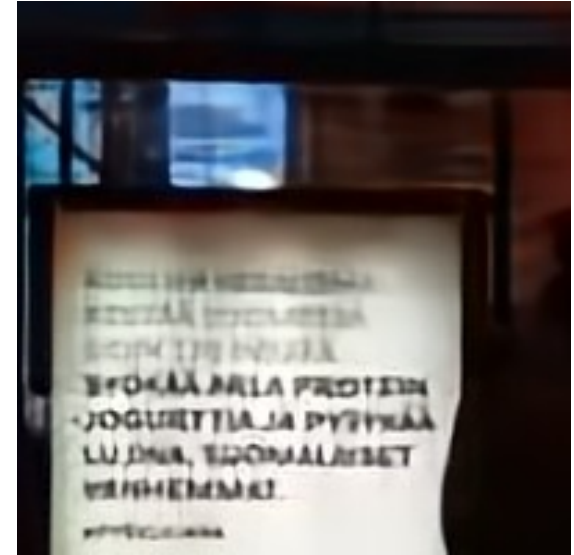
Text distortion



Original



HM, 585 times



JPEG AI 3.2, 685 times

Text distortion



Original



HM, 86.3 times



JPEG AI 4.1, 87.9 times

Text distortion



Original



HM, 93.3 times

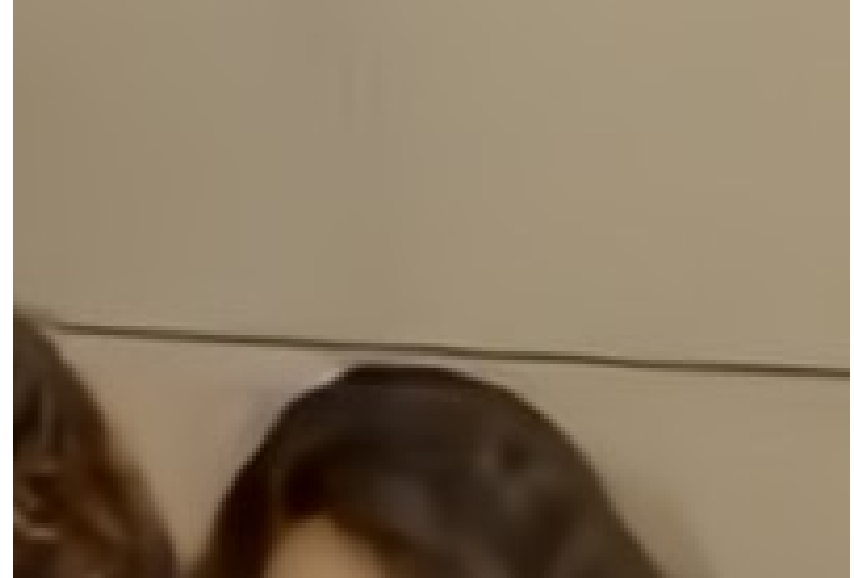


JPEG AI 4.1, 80.9 times

Edges distortion



HM, 285 times

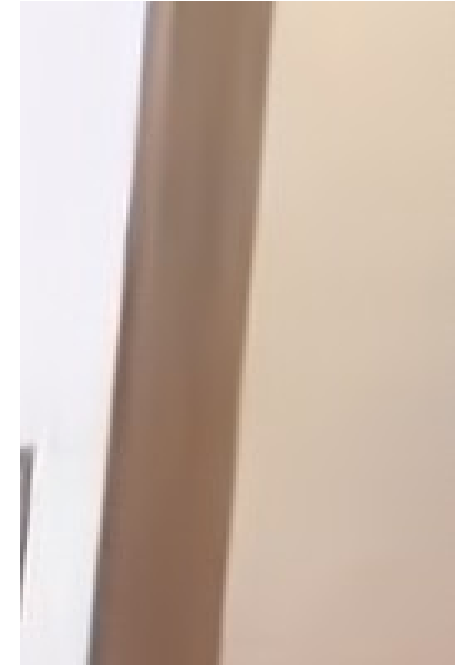


JPEG AI 3.2, 348 times

Edges distortion



HM, 381 times



JPEG AI 3.2, 500 times

Image corrupted by JPEG AI

VTM, 38.6 times



JPEG AI, 32.8 times

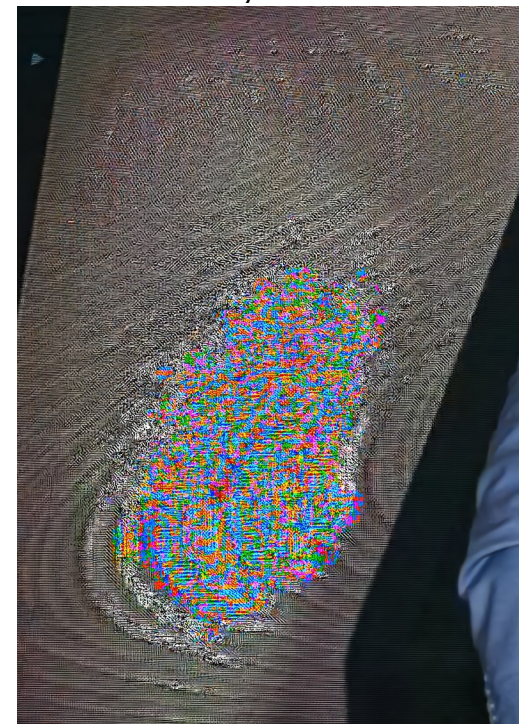
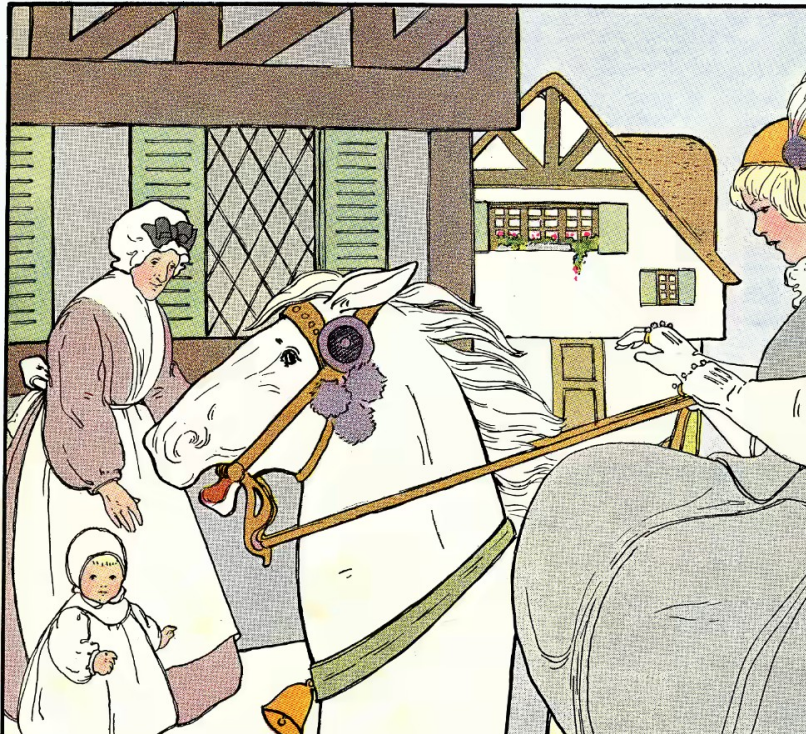


Image corrupted by JPEG AI

VTM, 27.7 times



JPEG AI, 29.1 times



Image corrupted by JPEG AI

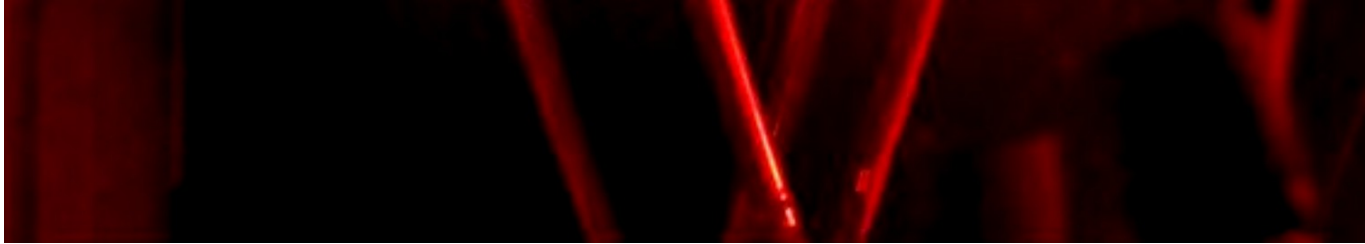
HM, 6.6 times



JPEG AI 3.2, 8.0 times



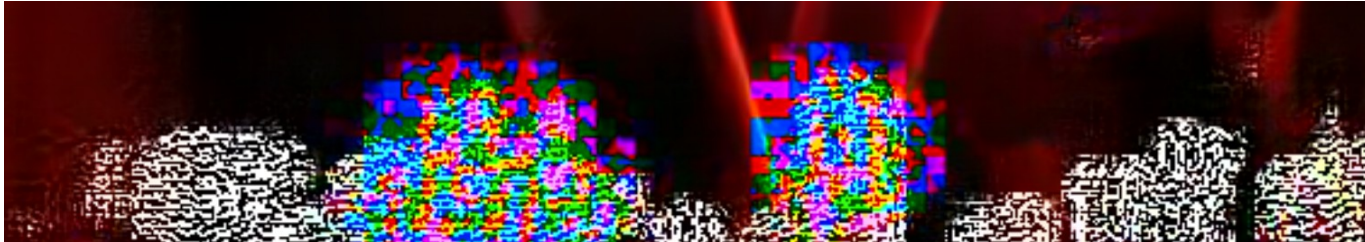
Image corrupted by JPEG AI



Original



HM, 172 times



JPEG AI 3.2
143 times

Comparison JPEG AI 3.2 vs 4.1

Version 3.2 vs 4.1

JPEG AI v3.2, 352 times



JPEG AI v4.1, 358 times

Version 3.2 vs 4.1

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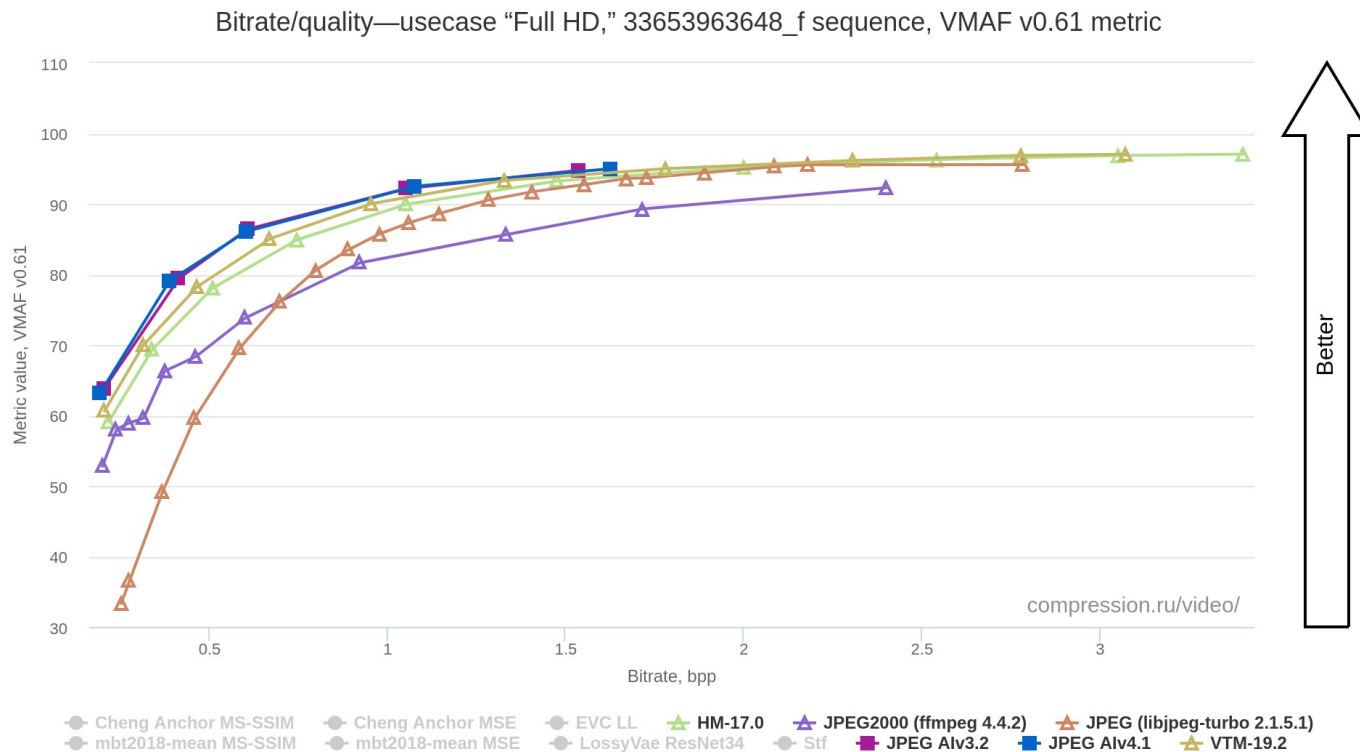
JPEG AI 3.2

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EDWARD K GARDING · EDWARD L GIBSON · L MARSHALL GIBSON · JOHN G GOTT

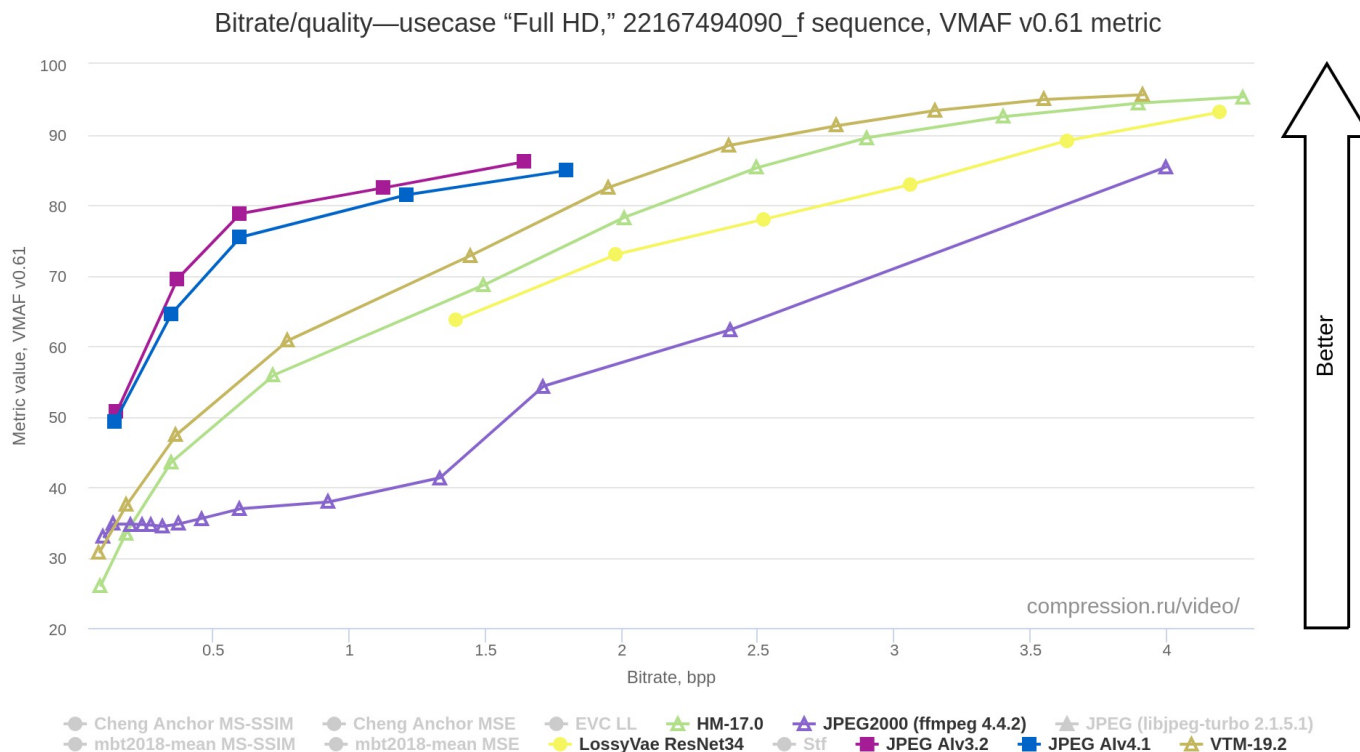
JPEG AI 4.1

Neural Image Compression Benchmark

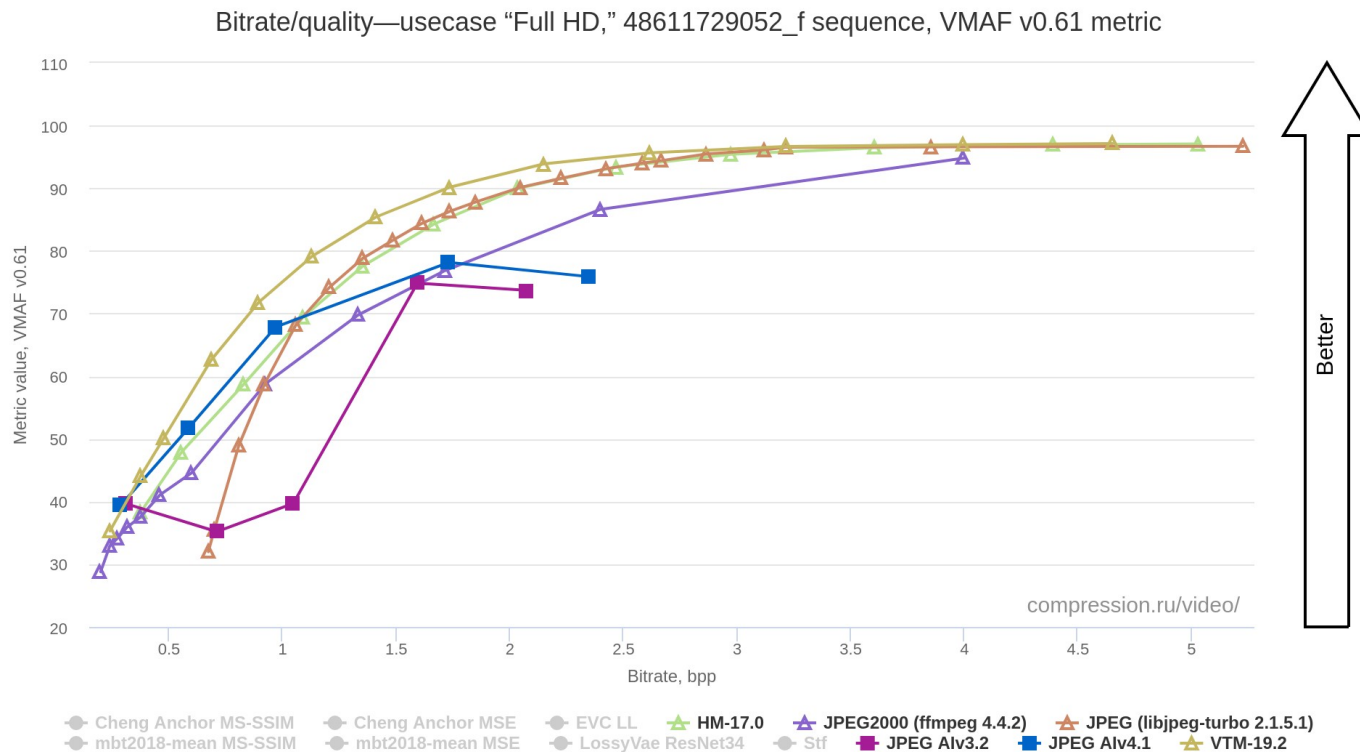
RD-curve example



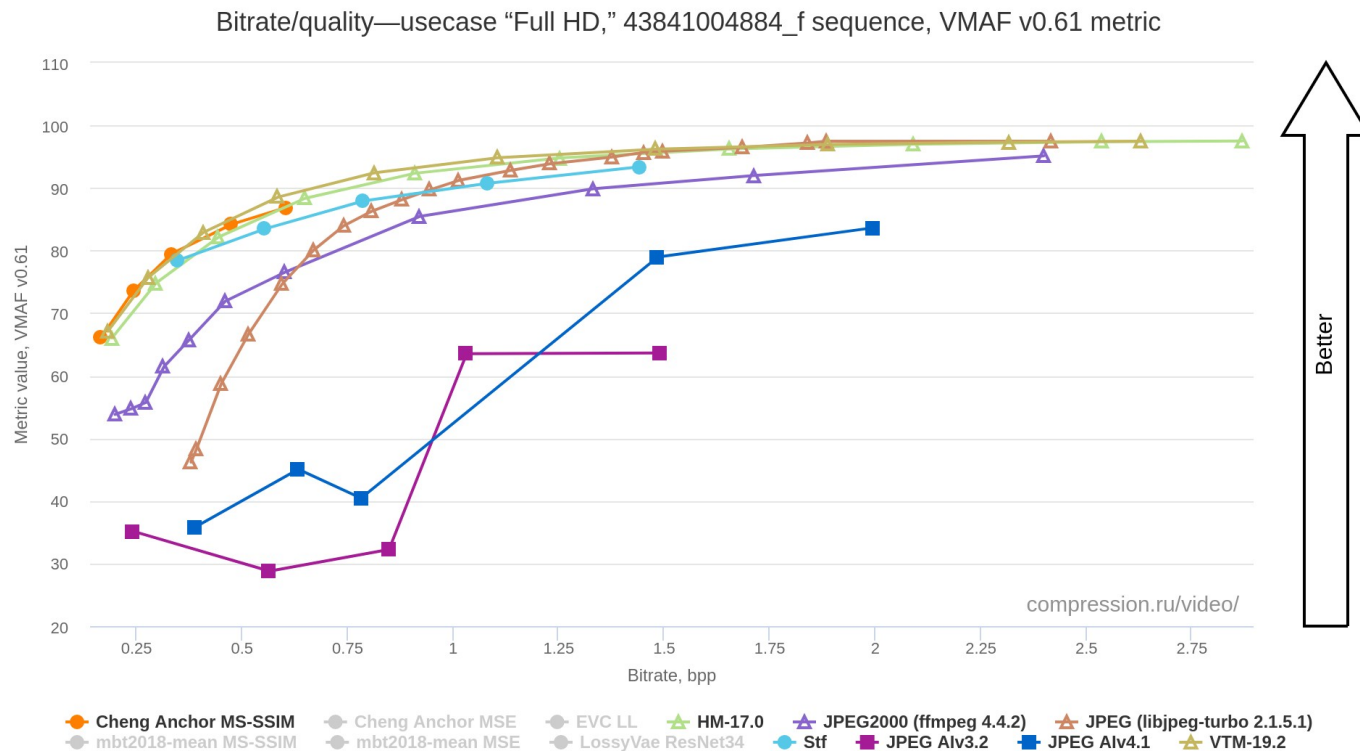
RD-curve example



RD-curve example



RD-curve example



Conclusions

- The topic of neural-network-based image compression has become extremely popular – neural codecs have surpassed traditional codecs, including VVC, by 30% in compression ratio over the past 5 years
- Textures with a regular structure can be completely distorted
- “Additionally drawn” borders and new objects may appear on the texture
- Existing image and video quality assessment methods are worse at assessing neural compression artifacts

Roadmap

- Decreasing artefact-sample cost (now)
 - Decreasing (Toloka \$)/(final sample)
 - Decreasing (computational costs, min)/(final sample)
- Dataset with 100.000 JPEG AI artifacts (~Dec'23)
- Dataset with 1.000.000 JPEG AI artifacts (dep. funding)
 - more artifact types
 - $\Rightarrow$ new metrics design
 - $\Rightarrow$ deeper SOTA methods/libraries testing
- MSU Image Compression Benchmark (~Feb'24)
- MSU Neural Compression Metrics Benchmark (~Sep'24)

Mid-term Goals

Our 2024 year goals:

- Benchmarking neural image/video compression methods
- Improving image/video quality metrics aimed at artifact detection
- Testing and improving the quality of JPEG AI libraries

Key ideas

- It's good idea to test product **BEFORE** release (not **AFTER**)
- **New metrics will help raise quality**
- Looks like **it's possible to improve subjective quality of JPEG AI** using Robustness Metric Benchmark methodology (less metric tuning during learning, higher correlation with human for better metrics)
- We found that **JPEG AI loses to VVC in our tests** (it's reasonable to compare testing methodologies in detail)

Similar artifacts in Super-Resolution

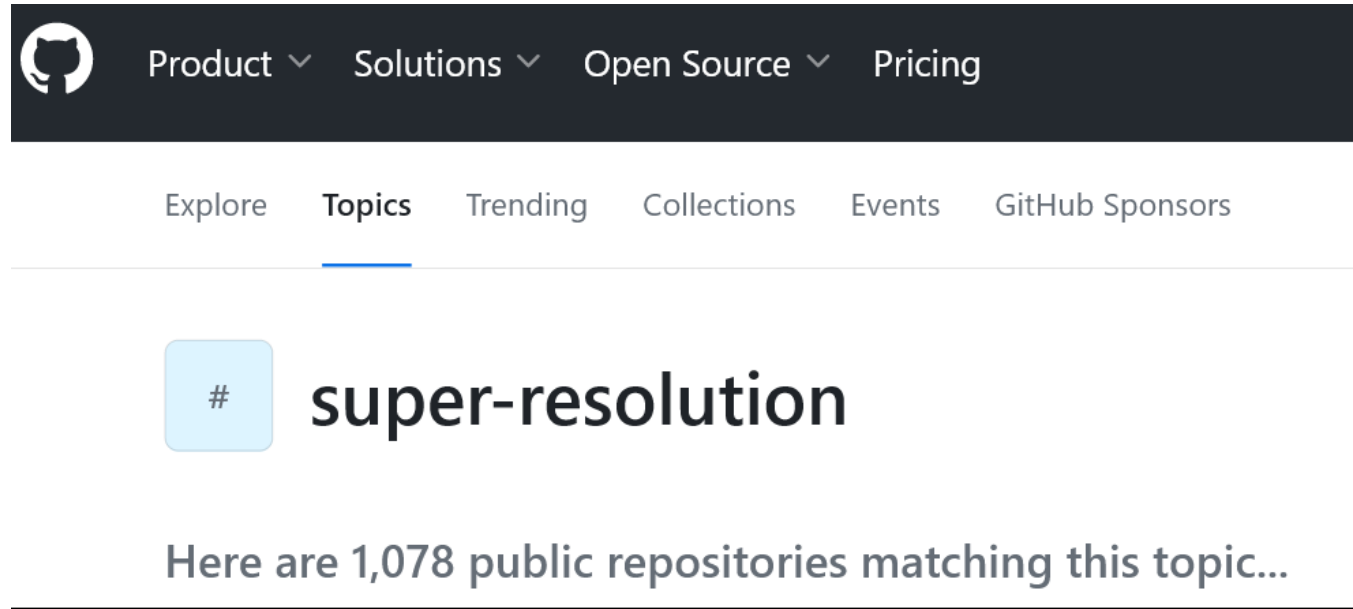
Introduction

- Resolution upscaling is the task of restoring a high-resolution image or video from a low-resolution counterpart
- Classic interpolation is inferior to neural-network-based algorithms in visual quality



Let's select the best SR

- Classical interpolation-based approaches are inferior to neural-network-based algorithms in visual quality
- 1078+ repositories <https://github.com/topics/super-resolution> (!)



Object distortion

GT ERQA=1.0	DIVSR ERQA=0.916	TMNet ERQA=0.891	BasicSR ERQA=0.852	TDAN ERQA=0.835	iSraBetter ERQA=0.786	RBPV ERQA=0.782	DynaVSR-V ERQA=0.651
DUF-28L ERQA=0.667	DynaVSR-R ERQA=0.669	RSDN ERQA=0.68	SOF-VSR-BB ERQA=0.695	SOF-VSR-BI ERQA=0.732	TGA ERQA=0.734	LGFM ERQA=0.751	D3Dnet ERQA=0.645
RRN-16L ERQA=0.629	DUF-16L ERQA=0.614	HCFlow ERQA=0.613	ESRGAN ERQA=0.599	waifu2x-cunet ERQA=0.593	GFPGAN ERQA=0.581	Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571
Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571	Real-ESRGAN ERQA=0.571

Results of SR methods on the text

Faces distortion



Bicubic



SwinIR

Faces distortion



Real-ESRGAN

Text and color distortion



Bicubic



RealSR

Edges distortion

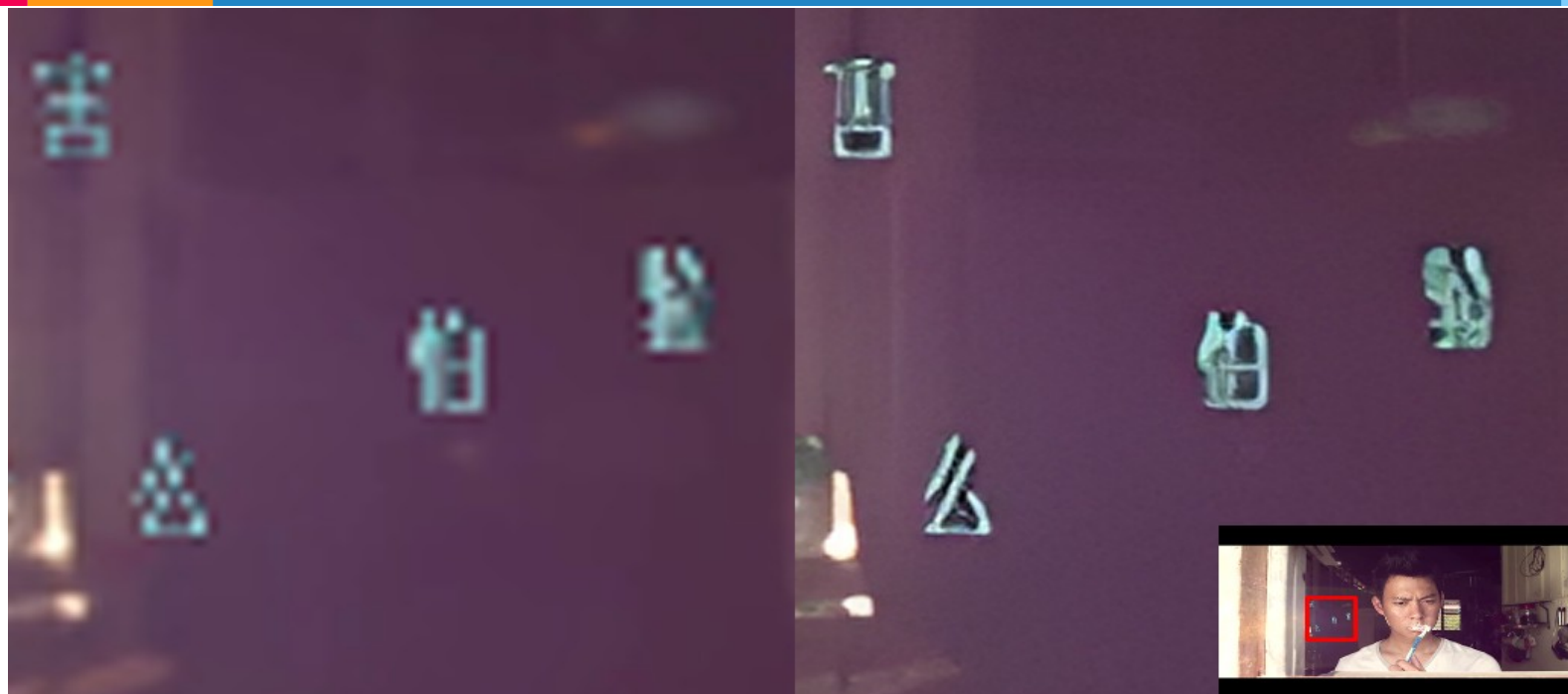


Bicubic



ESRGAN

Object distortion

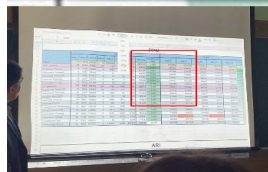


Bicubic

Real-ESRGAN

Object distortion

FFV1						FFV1					
7	Total size	Compression ratio	test_8	test_9		7	Total size	Compression ratio	test_8	test_9	
000	3239412	1.000	2152069	3725728		000	3239412	1.000	2152069	3725728	
019	124562	0.038	487253	1148917		019	124562	0.038	487253	1148917	
021	151315	0.047	421384	1344398		021	151315	0.047	421384	1344398	
201	146395	0.045	404024	1378836		201	146395	0.045	404024	1378836	
819	141905	0.044	434543	1432307		819	141905	0.044	434543	1432307	
866	139496	0.043	434537	1424388		866	139496	0.043	434537	1424388	
556	141614	0.044	434768	1438632		556	141614	0.044	434768	1438632	
967	148114	0.046	585291	1337800		967	148114	0.046	585291	1337800	
272	144830	0.045	515136	1479977		272	144830	0.045	515136	1479977	
			454749	1640706		870	151058	0.047	454749	1640706	
			471609	1648928		602	160011	0.052	471609	1648928	
			478159	1668157		391	170667	0.063	478159	1668157	
			533851	1827333		213	181638	0.058	533851	1827333	
						885	206667	0.067	561221	2006388	



Bicubic

Real-ESRGAN

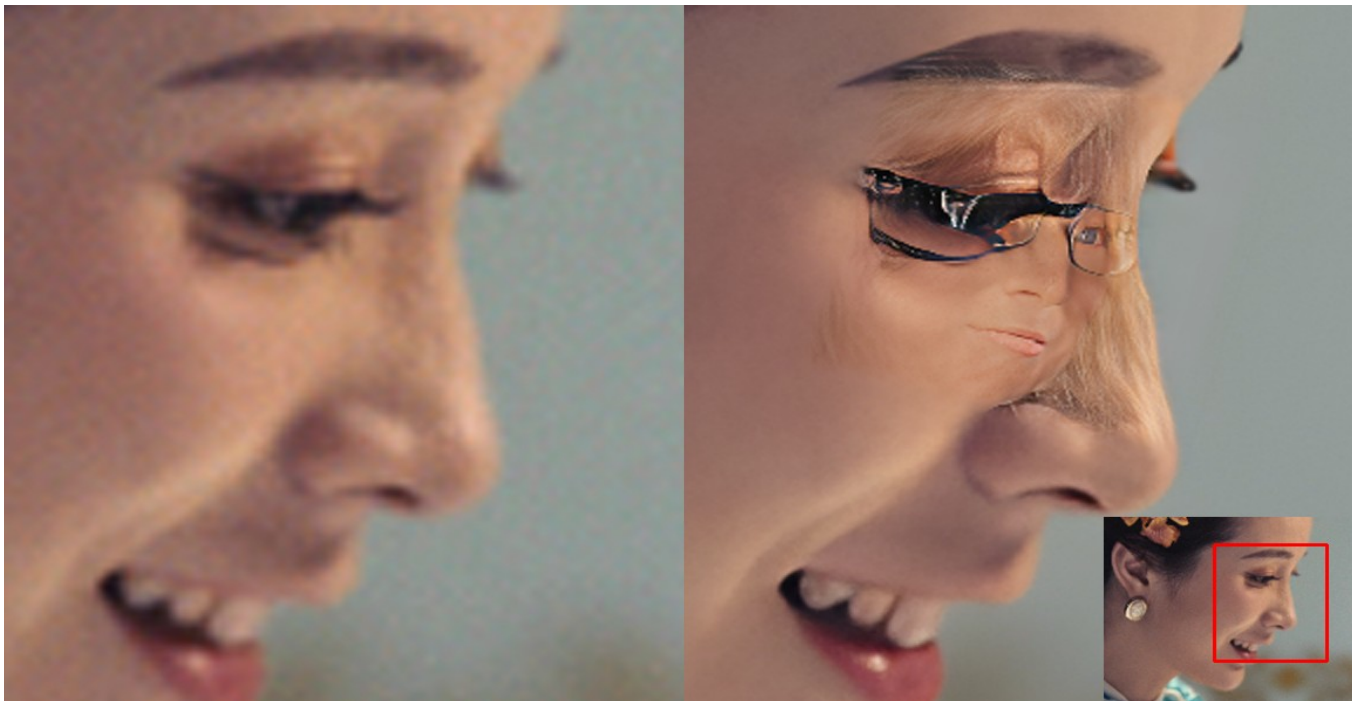
Adding objects



Bicubic

GFPGAN

Adding objects



Bicubic

GFPGAN

Adding objects



Bicubic

GFPGAN

Texture change



Bicubic

RealBasicVSR

Texture change



Bicubic

Real-ESRGAN

Style change



Bicubic



RealSR

Style change



Bicubic

Topaz Video Enhance AI AMQS 1

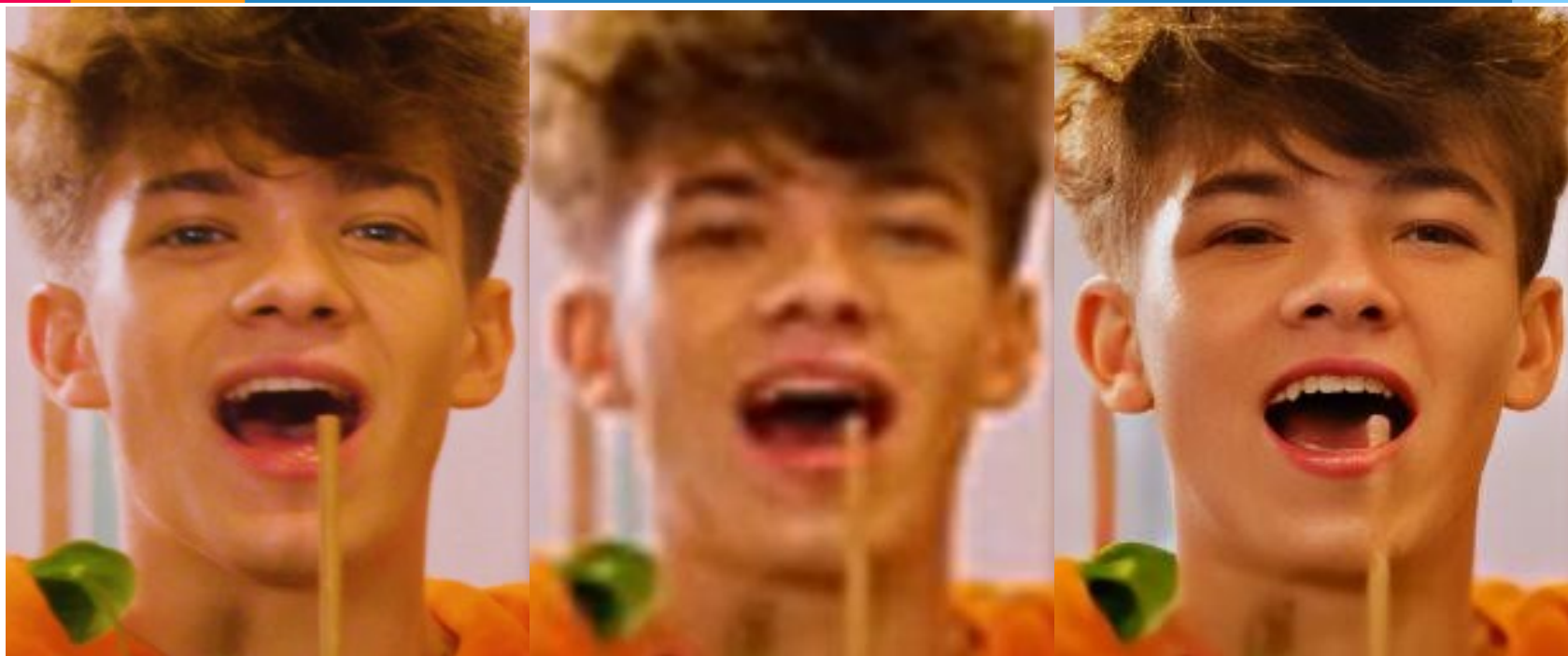
Face distortion



Bicubic

BSRGAN

Face distortion



Source

Bicubic

GFPGAN

Partial change of sharpness



Bicubic



SwinIR

Object distortion

Often videos are processed using image SR methods, applying them frame by frame



RealSR

Go deeper on HABR



3DVideo 14 фев в 10:00



Увеличь это! Современное увеличение разрешения в 2023



Средний



26 мин



17K

Алгоритмы*, Обработка изображений*, Машинное обучение*, Научно-популярное, Искусственный интеллект

Обзор



Бюст Зевса, увеличение разрешения в 4 раза

Edges distortion



Bicubic



RealSR

Video corruption



DynaVSR-R

Conclusion

- Distortion and addition of objects can seem unnatural and sometimes even frightening to people, so the subjective quality drops a lot
- Style changes are often invisible, especially when the original image is missing
- The main problems:
 - Face distortion
 - Regular texture distortion
 - Text distortion
- Many SR methods are not able to correctly restore compressed videos
- Detection of neural network artifacts is a non-trivial video processing task

Fixed Ultra Restoration SR



The idea

Recover resolution up to 4 times

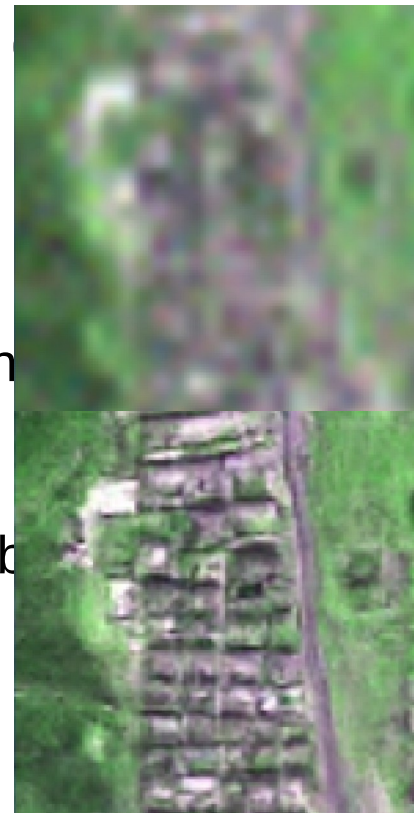


Input: 14+ uncompressed photos taken from one point

Output: reconstruction of the resolution
up to 4 times

Usage: improve resolution of landscape photos shot on
DSLR cameras or drones

Example: a drone with the resolution of one meter will be able to see
objects as small as 25cm



Our solution

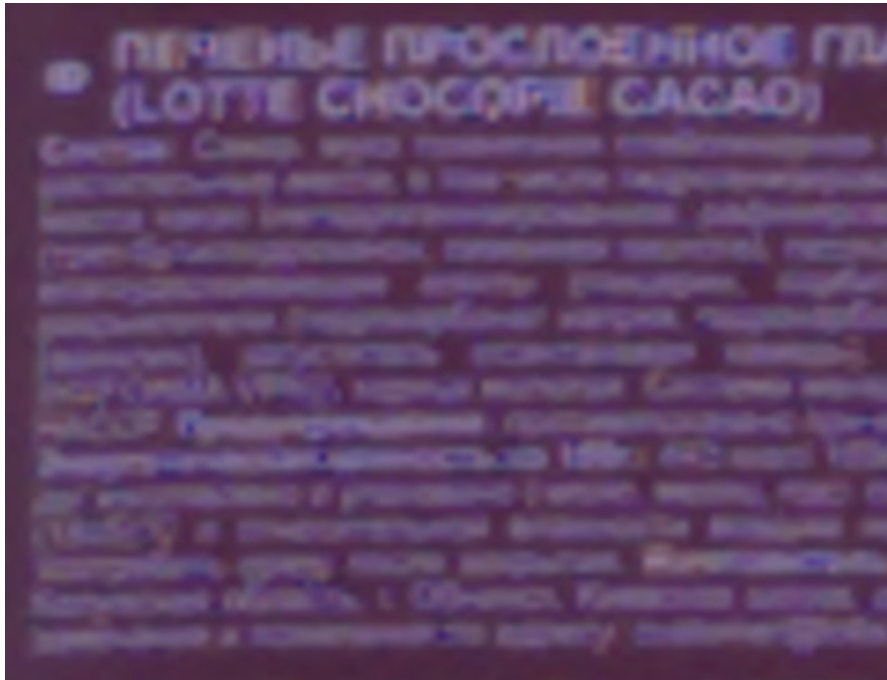


Before

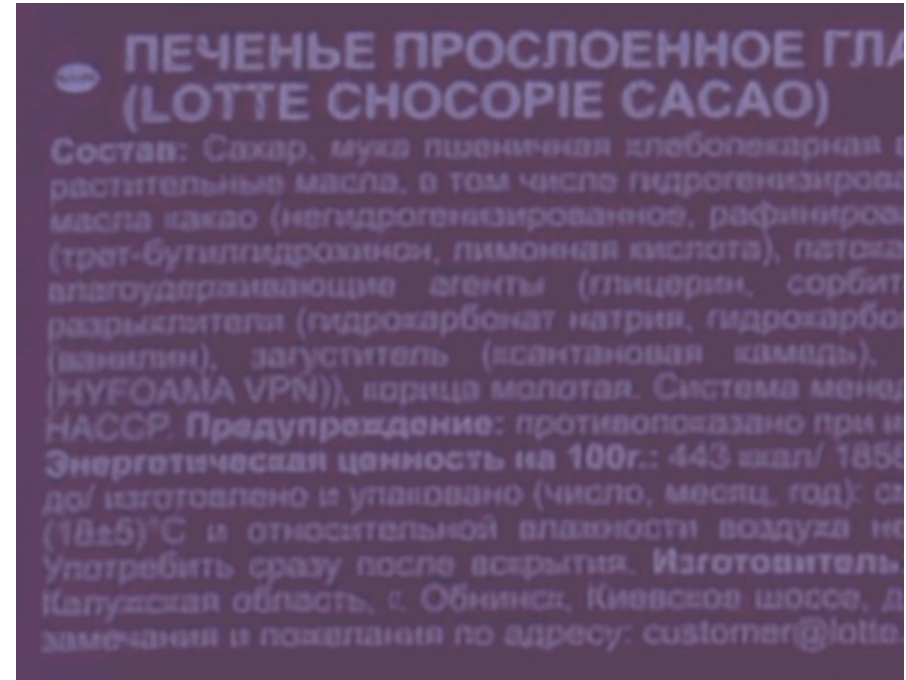


After

Our solution



Before



After

Our solution



Before



After

RealVSR++

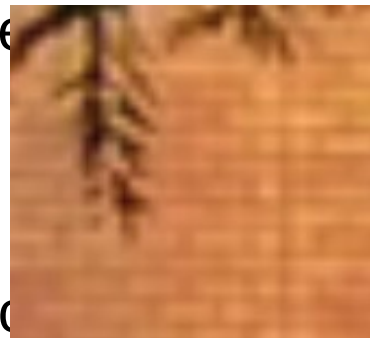


RealVSR++

Idea



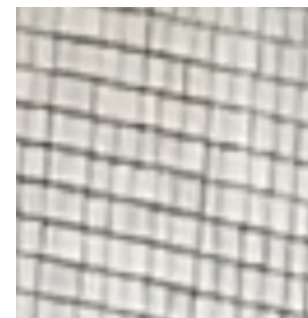
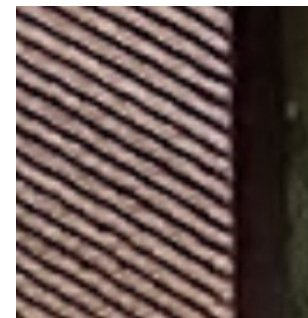
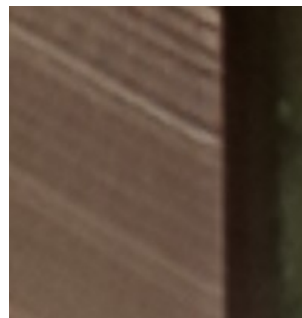
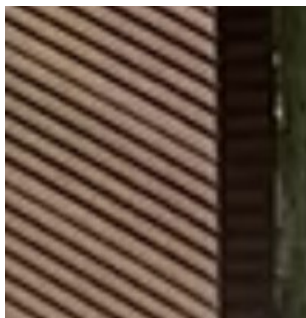
- Problem: almost all modern SR-models are trained on synthetic data.
- Therefore these models are not able to deal well with images captured on camera.
- Solution: collect a dataset of LQ-HQ sequences of real camera and train SR based on this data.
- Idea: **we can make any SR better!**



Real-world LQ/HQ pair

RealVSR++

Improving quality with real-world data training



Real-world GT

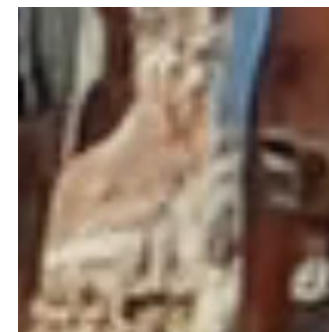
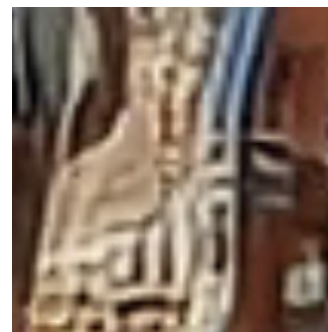
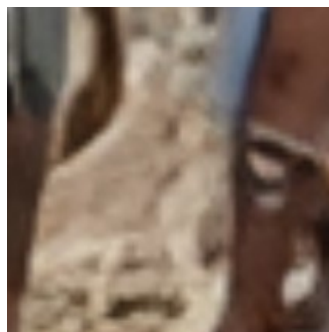
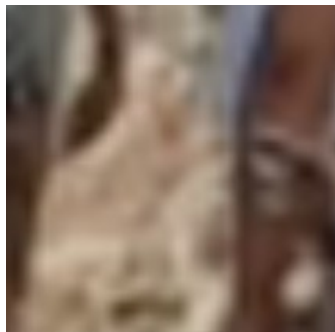
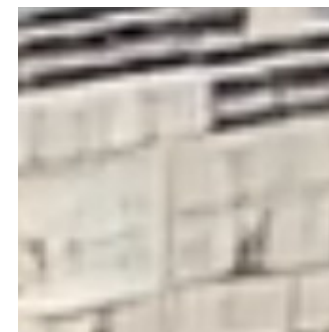
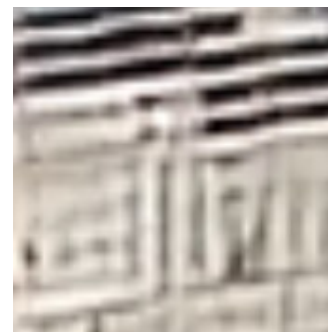
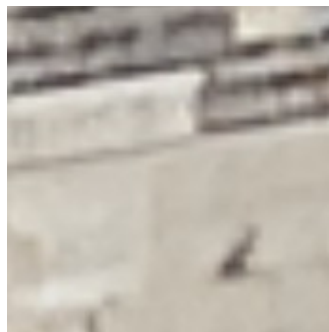
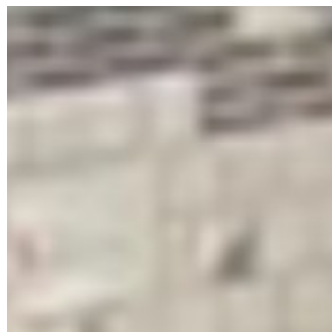
Real-world LQ

After applying trained
on synthetic data SR

After applying trained
on real-world data SR

RealVSR++

What ++ stands for



LQ (no-reference)

After applying trained
on synthetic data SR

After applying SR trained
on "bad" real-world data

After applying SR trained
on good real-world data

SR Quality Metrics Benchmark (Oct'23)

SR quality metrics

Poor traditional metrics performance



Artifacts caused by Super-Resolution cannot be detected by traditional metrics



Right video has better PSNR but worse visual quality (by subjective assessment)

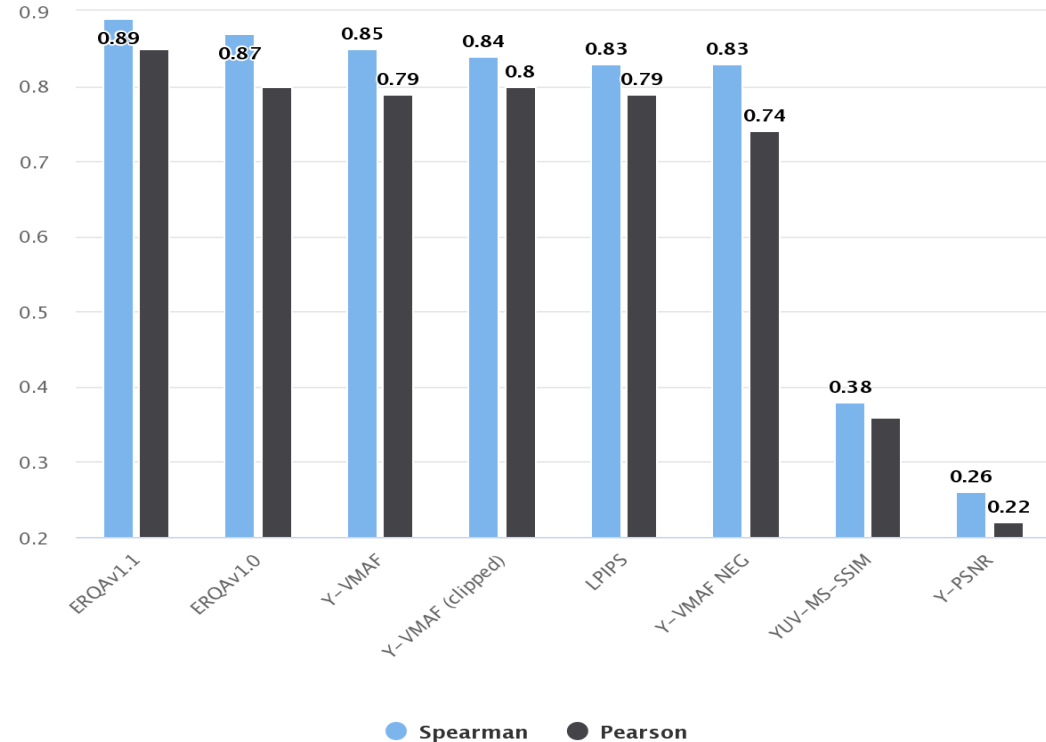
SR quality metrics

Correlations of metrics with subjective assessments



For subjective comparison:

- x264 codec
- 3 different bitrates (1000, 2000, 4000 kbps)
- 2 crops with resolution 320×270 from each video

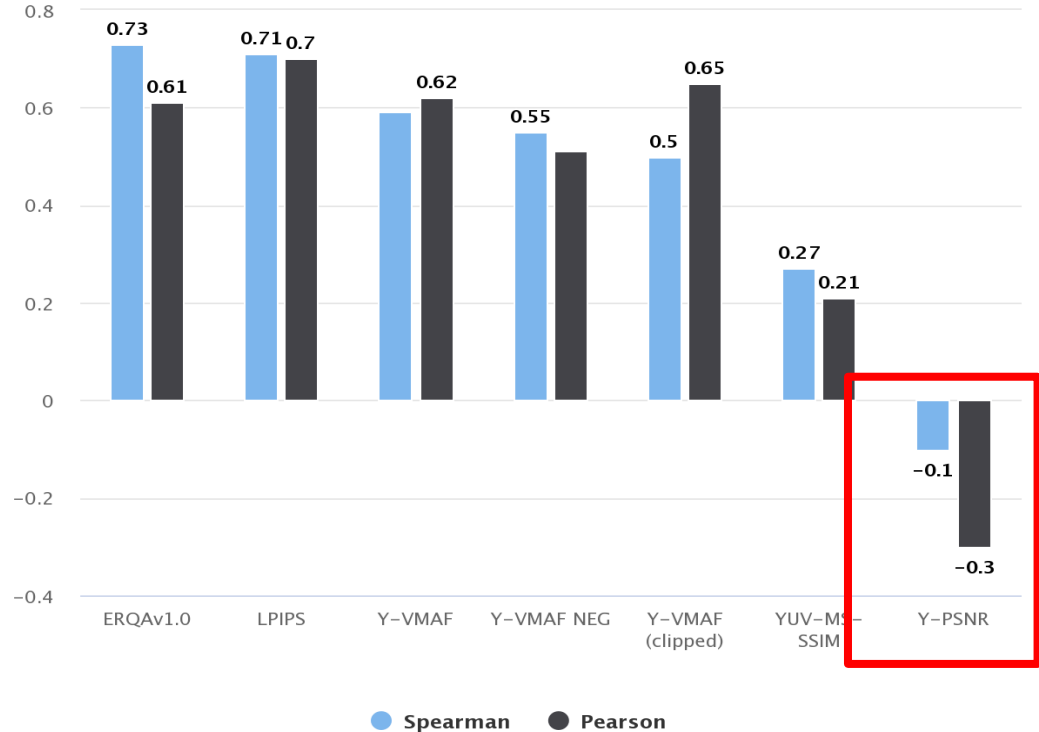


SR quality metrics

Correlations of metrics with subjective assessments

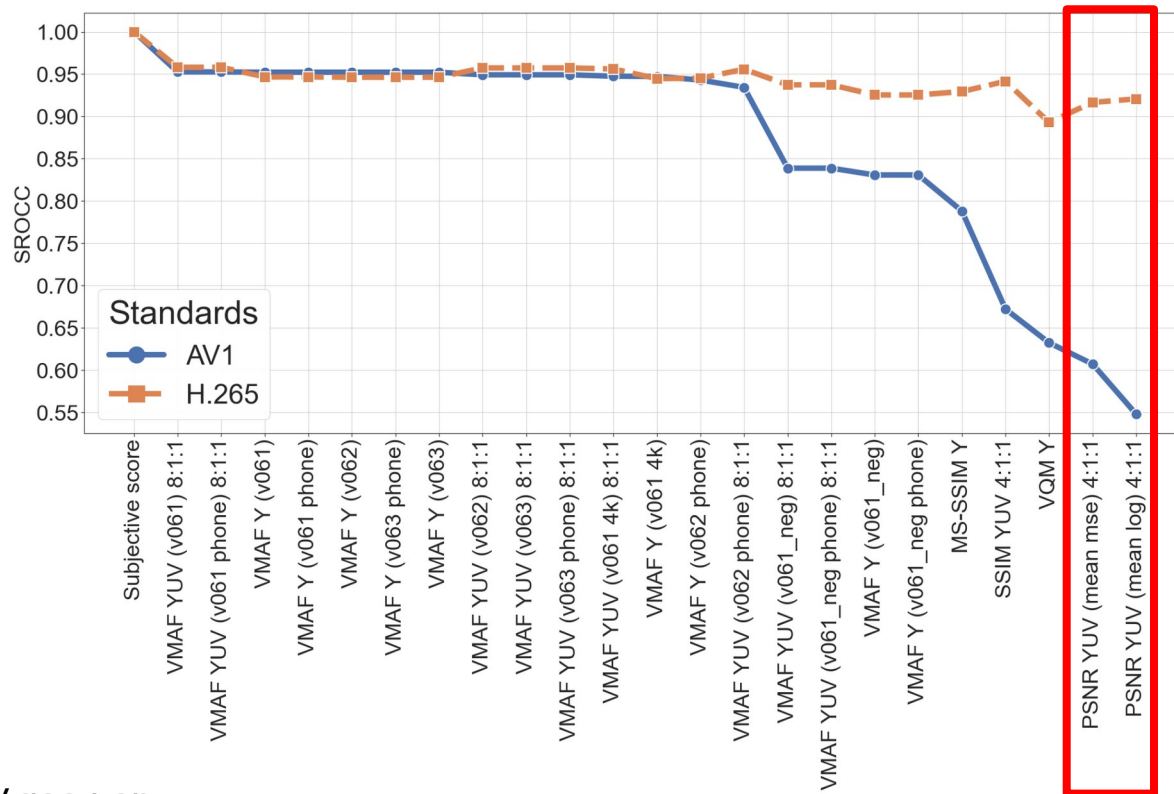


BSQ-rate was calculated via subjective results extrapolation using the most similar objective metric



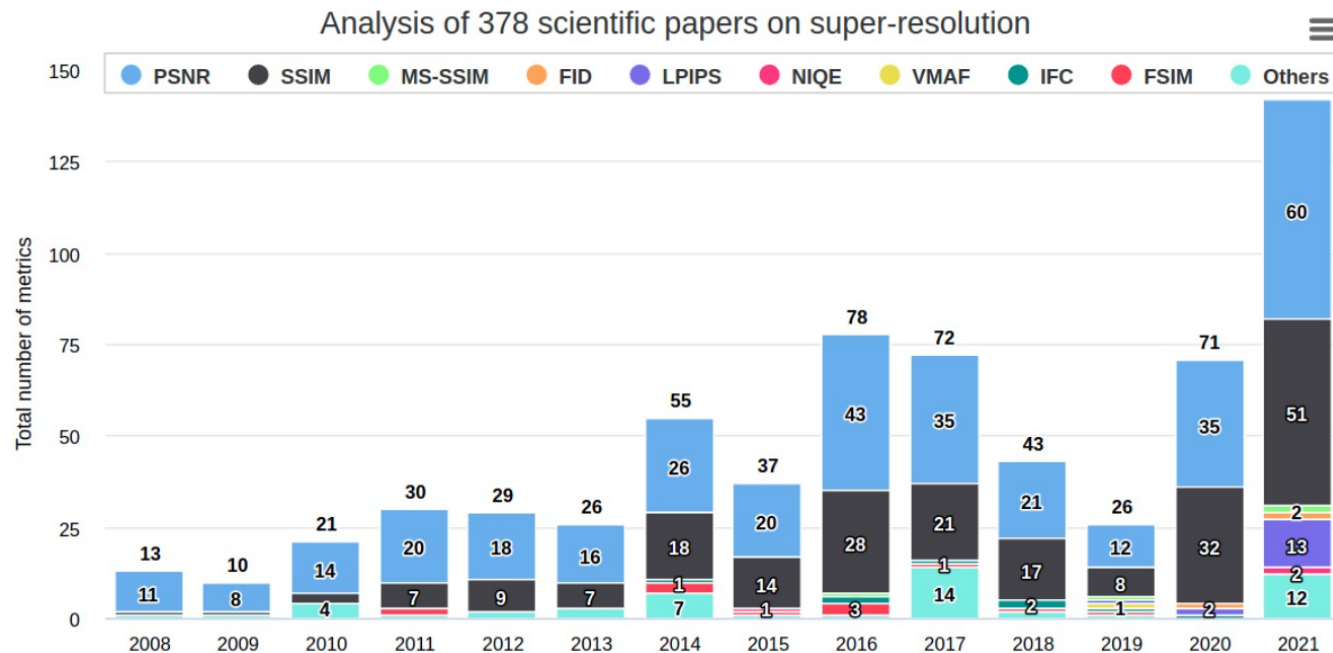
Video Quality Metrics Benchmark

Dramatic PSNR degradation on AV1 vs H.265



SR quality metrics

PSNR still very popular



Our paper “ERQA: Edge-Restoration Quality Assessment for Video Super-Resolution”

MSU SR Metrics Benchmark

Here you can find 46 Super Resolution metrics and choose the most suitable for your videos

Powered by
 & 
 G&M Lab head: Dr. Dmitry Vatolin
 Measurements, analysis:
 Artyom Borisov,
 Egor Kashkarov,
 Eugeny Bogatyrev

Metric	iSeeBetter	ggPXL VCOMP
PSNR	19.71	19.37
MDTVSFA	0.44	0.50
SUBJECTIVE SCORE	1.61	5.61

Do you know, that PSNR isn't the best metric?

Complex Content

- 1000+ distorted videos with different types of content
- 46 Super Resolution methods were used to distort videos
- RealSR, iSeeBetter, waifu2x, VRT and Real-ERGAN are among them.

A lot of Metrics!

- 46 Super Resolution Metrics for different tasks
- All of them were tested on 1000+ videos
- The list of metrics is **constantly updated**

Large Leaderboard

- **Useful charts**, that shows working time and results of metrics
- **Large Leaderboard** with 3 types of correlation between human's subjective scores and metrics' results

- **50+ VQA and IQA metrics**
- **Constantly updating** list of metrics
- Testing on **1000+ distorted videos** covering a lot of use cases

The main page of Benchmark

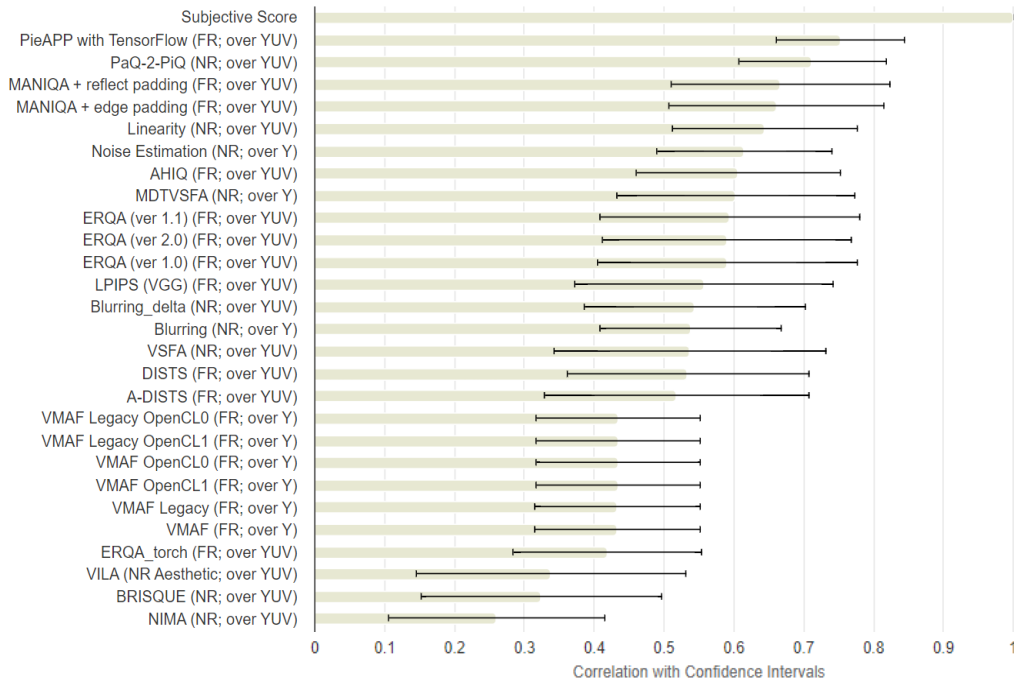
MSU SR Metrics Benchmark

- **Large dataset** with different types of content: from *real videos* to *dynamic ads* and *video games*
- Videos are preprocessed with 5 video codecs
- **46 Super-Resolution methods**
- **More than 100000 frames**
- Subjective comparison with more than **160000 responses**



Dataset Preview

MSU SR Metrics Benchmark



- **Large leaderboard of correlation with subjective score, comparing different metrics versions**
- **3 types of correlation: SROCC, PLCC and KLCC**
- **Runtime-to-quality leaderboard**

Current Leaderboard sorted by SROCC

MSU SR Metrics Benchmark

- The data is presented in the form of **plots and tables**
- **Speed/Quality Tradeoff** with Pareto-Optimal Front



SR Metrics Benchmark Conclusion

- Benchmark **will be published pretty soon** (Oct'23)!
- The nature of SR & JPEG AI artifacts is very similar, so **SR metrics will be used in development**
- We can get an earlier start on neural network artifacts **in video**

Contacts

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